INF10025 Tutorial 07 Answers

Complete Solutions & Explanations for Week 7

Question 1 & 2: COUNT(*) Output Rows

Q1: Returns **1 row** containing the total row count integer.
Q2: Returns **1 row** with value **0**.

Question 3: GROUP BY Gender

Returns **2 rows** (one for 'M' and one for 'F').

Question 4: Invalid Aggregation Statement

Error: `BIRTHCOUNTRY` is a non-aggregate column selected alongside `COUNT(*)` without a `GROUP BY BIRTHCOUNTRY` clause. Triggers `ORA-00937: not a single-group group function`.

Question 5: Useless GROUP BY Statement

`SELECT COUNT(*) FROM ACTOR GROUP BY BIRTHCOUNTRY` displays count numbers without selecting `BIRTHCOUNTRY`, rendering counts unidentifiable.

Question 6: Mismatched GROUP BY Column

Selecting `BIRTHCOUNTRY` while grouping by `GENDER` is invalid because `BIRTHCOUNTRY` is not in the `GROUP BY` clause.

Question 7: Multi-column GROUP BY Example Output

SELECT BIRTHCOUNTRY, GENDER, COUNT(*) FROM ACTOR GROUP BY BIRTHCOUNTRY, GENDER;
Sample Output:
Australia | M | 12
Australia | F | 15
USA | M | 45

Question 8 & 9: MIN() Aggregate Function

Q8: `MIN()` calculates the minimum/earliest value in a column across a group.
Q9: Returns **1 row** containing the earliest birthdate.

Question 10 - 15: STUDENT Table Queries

```
Q10 Total Students: SELECT COUNT(*) FROM STUDENT;
Q11 Female Students: SELECT COUNT(*) FROM STUDENT WHERE Gender = 'F';
Q12 Students by Gender: SELECT Gender, COUNT(*) FROM STUDENT GROUP BY Gender;
Q13 Students by Degree: SELECT Degree, COUNT(*) FROM STUDENT GROUP BY Degree;
Q14 Degree Totals >= 2:
SELECT Degree, COUNT(*)
FROM STUDENT
GROUP BY Degree
```

```
HAVING COUNT(*) >= 2;
Q15 Female Avg Passed >= 3:
SELECT Degree, AVG(SubjectsPassed)
FROM STUDENT
WHERE Gender = 'F'
GROUP BY Degree
HAVING AVG(SubjectsPassed) >= 3;
```

Question 16: ALLOCATION ERD & Schema

a. Identifier: `EmpNo` + `TaskNo` (Composite Primary Key)
b. Foreign Keys: 2 (`EmpNo`, `TaskNo`)
c. Relational Schema:
EMPLOYEE(EmpNo, EmployeeName)
TASK(TaskNo, Description)
ALLOCATION(EmpNo, TaskNo)
FK (EmpNo) References EMPLOYEE, FK (TaskNo) References TASK
d. SQL Statements:

```
INSERT INTO EMPLOYEE VALUES (101, 'Jim Black');
INSERT INTO EMPLOYEE VALUES (102, 'Perry White');
INSERT INTO TASK VALUES (1, 'Bake Cake');
INSERT INTO TASK VALUES (2, 'Chop Onions');
INSERT INTO ALLOCATION VALUES (101, 2);
INSERT INTO ALLOCATION VALUES (102, 1);
```

Question 17: ATTACHED ERD & Schema

a. Identifier: `WID` + `JobName` (Composite Primary Key)
b. Foreign Keys: 2 (`WID`, `JobName`)
c. Relational Schema:
WORKER(WID, WName)
JOB(JobName, Description)
ATTACHED(WID, JobName, AgreedRate, DateNegotiated)
FK (WID) References WORKER, FK (JobName) References JOB

Question 18: HOUSE ERD & Schema

a. Identifiers:
SUBURB: `SuburbName`
BUILDER: `BuilderCode`
HOUSE: `SuburbName` + `StreetName` + `HouseNo`
b. Relational Schema:
SUBURB(SuburbName, Population)
BUILDER(BuilderCode, BuilderName)
HOUSE(SuburbName, StreetName, HouseNo, DateBuilt, *BuilderCode*)
FK (SuburbName) References SUBURB, FK (BuilderCode) References BUILDER